Alexandre Fournier Montgieux

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Technical Skills

Core Languages: Python, C++, Rust, C, SQL, OCaml, Java, GLSL, Bash
Deep Learning & ML: PyTorch, TorchLightning, Hugging Face, TorchVision, TensorFlow, Optuna, Scikit-learn, SciPy, NumPy
HPC & Systems Infrastructure: SLURM Workload Manager, Jean Zay (Nvidia A100/V100), Adastra (AMD ROCm MI250X), Factory-IA, Git, Linux/UNIX
Architectures & Paradigms: Vision Transformers (ViT), ResNet, iResNet, Flow Matching, (Latent) Diffusion Models, UNet, VAE, GANs, StyleGAN, Self-Supervised Learning, Contrastive Learning, 3D Object Detection

Experience

Doctoral Researcher in Machine Learning & Computer Vision — Palaiseau, France
CEA LIST – Nano-INNOV (Université Paris-Saclay) — *Nov. 2023 – Nov. 2026 (Ongoing)*

- **Distributed Vision Training at Scale:** Trained deep representation backbones (iResNet-50, ResNet-101, ViT-S; 50M to 100M parameters) on up to 2M+ image datasets using PyTorch Lightning on multi-GPU configurations across heterogeneous national supercomputers (Jean Zay / Nvidia, Adastra / AMD ROCm, and Factory-IA).
- **Multi-Node Hyperparameter Scheduling:** Orchestrated distributed efficient multi-node hyperparameter searches (for training deep backbones and inference pipelines) using Optuna with custom parallel database coordination.
- **Self-Supervised Representation Learning:** Implemented multi-GPU contrastive pre-training pipelines (e.g. SimCLR) for facial feature extraction, optimizing distributed negative sample batching and convergence stability.
- **Web-Scale Data Curation:** Processed and filtered the WebFace42M dataset (42 million images) with high-throughput streaming pipelines to generate diversified training datasets (Under Submission).
- **Synthetic Training Set Generation for Face Recognition:** Generated over 1M diverse synthetic face images using GAN and Diffusion Models. The resulting synthetic training set can be found here (WACV 2025).
- **Identity Anonymization via Latent Morphing:** Scaled diffusion-based latent trajectory morphing (DiffMorpher) and explored text driven generation to synthesize hundreds of thousands of identity-blended faces with anonymization guarantees for privacy-compliant benchmarking (WACV 2027).
- **Continuous Generative Modeling:** Co-developed Diff-CA (NeurIPS Spotlight), training continuous flow matching diffusion models to cleanly separate shared visual factors from salient characteristics.
- **Teacher-Student 3D Perception:** Implemented a weakly supervised multi-view teacher-student knowledge distillation pipeline for 3D object detection in autonomous driving benchmarks (WACV 2026).

Referent Teacher – Advanced Deep Learning, Master BDMA — Gif-sur-Yvette, France
CentraleSupélec — *2024 – 2026*

- Designed and instructed **master-level (M2) coursework** covering deep computer vision architectures, optimization methods, detection and segmentation techniques, and auditing methodologies.

Research Intern – Generative Modeling with GANs — Palaiseau, France
CEA LIST – Nano-INNOV — *Apr. 2023 – Sep. 2023*

- Engineered **generative adversarial networks** (StyleGAN) to synthesize balanced visual training datasets, achieving empirical reductions in verification disparities across benchmark datasets.

Computer Vision Intern – Automated Lithology Characterization — Clamart, France
Schlumberger / SLB — *Feb. 2022 – Aug. 2022*

- Developed **segmentation approaches** and **frugal classification models** to automate rock grain size estimation from high-resolution 2D microscopic imagery.

Front-End & Platform Developer – BackOffice Team — Paris, France
Stockly — *Aug. 2021 – Feb. 2022*

- Participated in the creation and development of the **deployed BackOffice** platform (coded in Rust and Angular). The platform aimed to improve the company's logistics management, bringing order and supplier workflows into a single working environment alongside existing systems.

Software & Systems Engineering

REGI-Face Toolkit | *Integrated in European Vanguard Project* 2026
- Engineered and packaged a modular Python library for automated, frugal demographic inference for face analysis; integrated into the European Vanguard project for robust, high-throughput facial data processing.

FaVFA (Fairness in Face Verification Analysis) | *Open Source & APP Software Deposit* 2025
- Designed and open-sourced an end-to-end framework for statistical bias evaluation in face recognition models (WACV 2025). Formally registered with the Agence pour la Protection des Programmes (APP Deposit N° IDDN.FR.001.160022.000.S.P.2025.000.31235).

Slurm Orchestration Framework | *Internal HPC Computing Utility* 2024 – 2026
- Developed an automated Python utility to standardize and generate modular Slurm batch execution scripts, handling multi-GPU single-node launches, environment isolation, and logging on Jean Zay, Adastra, and Factory-IA.

High-Performance Sparse Matrix Optimization | *Large-Scale Recommendation Systems* 2025 – 2026
- Identified memory and runtime bottlenecks in cross-domain user coherence estimation (cf. WWW 2026); patched internal SciPy sparse matrix routines to compute ultra-large sparse graphs under constrained RAM limits.

Publications

Portrait Fitness as a Pretext: Probing Implicit Demographic Bias in VLMs WACV 2027
- **Fournier-Montgieux, A.**, Le Borgne, H., Popescu, A., & Luvison, B.

Diff-CA: Separating Common and Salient Factors with Diffusion Models NeurIPS 2026 (Spotlight)
- Soumm, M., **Fournier-Montgieux, A.**, He, Y., Gori, P., & Newson, A.

Quantifying User Coherence: A Unified Framework for Cross-Domain Recommendation Analysis WWW 2026
- **Fournier-Montgieux, A.**, Soumm, M., Popescu, A., & Delezoide, B.

MVAT: Multi-View Aware Teacher for Weakly Supervised 3D Object Detection WACV 2026
- Lahlali, S., **Fournier-Montgieux, A.**, Granger, N., Le Borgne, H., & Pham, Q. C.

Reliable and Reproducible Demographic Inference for Fairness in Face Analysis arXiv 2025 / Under Review
- **Fournier-Montgieux, A.**, Le Borgne, H., Popescu, A., & Luvison, B.

Fairer Analysis and Demographically Balanced Face Generation for Fairer Face Verification WACV 2025
- **Fournier-Montgieux, A.**, Soumm, M., Popescu, A., Luvison, B., & Le Borgne, H.

Research Projects

Top Reviewer | Neurips 2026
SHARP Project | Inter-lab Collaboration 2023 – 2026
Research Visit | CERTH, Thessaloniki, Greece April – May 2024

Education

MSc in Mathematics, Vision, and Learning (MVA) Cachan, France
ENS Paris-Saclay – Mention: Très Bien (Highest Honors) *2022 – 2023*
- Completed an advanced curriculum spanning deep learning, computer vision, NLP, reinforcement learning, generative models, and convex optimization. Key projects include:
  - Adapted open-weight LLMs (LLaMA/XLMRoBERTa) and specialized architectures for multilingual Named Entity Recognition (NER) on low-resource African languages (extending the MasakhaNER benchmark).
  - Formulated and evaluated Denoising Diffusion Probabilistic Models (DDPM) adapted to image restoration and non-linear dequantization inverse problems.
  - Implemented and benchmarked Plug-and-Play (PnP) proximal optimization algorithms (PnP-α PGD, ADMM, LADMM) for image restoration under spatially variable blur.

Engineering Degree (Diplôme d'Ingénieur) in Computer Science & Machine Learning Paris, France
Télécom Paris – GPA: 4.0 / 4.0 *2019 – 2023*
- Cursus focused on Image processing, Deep Learning, and Human-Machine Interaction. Project examples include:
  - Encoded GPU accelerated algorithms (OpenCL) such as bayesian super-resolution algorithm in the Fourier Domain, and particle-based fluid simulation algorithm.
  - Co-developed a decentralized React-Native chat application based on the Tangle (alternative to blockchain), details here

Classes Préparatoires aux Grandes Écoles (MPSI / MP*) Paris, France
Lycée Janson de Sailly *2017 – 2019*

Languages

Languages: French (Native), English (C1/Fluent), German (C1/Working Proficiency)